Paavan Shah

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Summary

Robotics and embedded firmware engineer with practical experience in autonomous navigation, real-time control, multi-sensor embedded systems, and scientific instrumentation. Strong background in ROS2-based robotic systems, hardware debugging, communication protocols (SPI/I2C/UART), and C/C++ firmware development. Experienced in developing complete systems including autonomous rovers, near-space payloads, CubeSat prototypes, and embedded sensing platforms. Interested in robotics, space systems, embedded firmware, and low-level system design.

Core Skills

Programming: Embedded C, C++, Python, MATLAB, ROS2, Verilog

Microcontrollers: ARM Cortex-M, Arduino, Jetson Nano, Raspberry Pi, Pixhawk

Protocols: SPI, I2C, UART, CAN, LoRa, RS485

Tools: Git, Oscilloscope, Logic Analyzer, JTAG, Ubuntu/Linux, Arduino IDE, MATLAB, EasyEDA, Xilinx Vivado, Fusion 360

Domains: Robotics Firmware, Autonomous Navigation, Satellite Systems, Wireless Communication, Embedded Systems

Experience

Graduate Engineer Trainee (GET), Zydus MedTech – Ahmedabad Jun 2026 – Present

- Working in the HemoD Electrical Department on electrical and electronic systems for medical-device applications.
- Supporting sensor evaluation, hardware testing, technical investigation, and system-level validation activities.
- Contributing to proof-of-concept development and evaluation of electronic subsystems.

Research Intern, Physical Research Laboratory (PRL) – Ahmedabad Jan 2026 – 12 Jun 2026

- Worked on atmospheric instrumentation and embedded data-acquisition systems for scientific experiments.
- Developed firmware and sensor-integration solutions for pressure, temperature, humidity, flow, and current measurements.
- Collaborated on the development and testing of an automated air-sampling payload for aerosol vertical profiling.

Space Tutor Intern, Origin Antares 1 May 2025 – 30 Nov 2025

- Provided online tutoring on satellite systems, CubeSat design, and space communication.
- Mentored 50+ students on practical aspects of satellite technologies and space systems.

Research Intern, Ural Federal University – Yekaterinburg, Russia 21 Jul 2025 – 3 Aug 2025

- Completed an international AI/ML internship, applying Python and machine-learning techniques to real-world datasets.
- Contributed to "NucLearn," a gamified educational platform integrating simulations, chatbot functionality, and interactive modules.
- Implemented AI logic during the HackAtom hackathon while collaborating with an international team.

Research Intern, Sardar Vallabhbhai National Institute of Technology (SVNIT) – Surat 26 May 2025 – 30 Jun 2025

- Developed a perception-to-planning pipeline for autonomous drone navigation.
- Implemented real-time APF and VFH obstacle-avoidance algorithms in Unreal Engine.
- Deployed the system on Jetson Nano with Intel RealSense and Open3D for embedded execution.

Junior Research Intern, Space Applications Centre (SAC), ISRO – Ahmedabad 31 May 2024 – 9 Aug 2024

- Conducted electro-optical characterization and data analysis of the SWIR6000 sensor.
- Produced 50+ performance graphs using LabVIEW for sensor data acquisition and characterization.

- Operated and troubleshot hardware setups for optical detectors and image sensors.

Projects

Automated Air Sampling System for Aerosol Vertical Profiling

- Designed and developed an automated atmospheric air-sampling system for aerosol vertical profiling.
- Developed embedded data-acquisition and control electronics integrating pressure, temperature, humidity, flow, and current sensing.
- Implemented pressure-aware sampling control to maintain consistent air collection under varying atmospheric conditions.

Junarakar Sat-01 | Near-Space Weather Satellite Project Project Demo

- Led the end-to-end design and development of a near-space atmospheric payload for real-time weather data acquisition.
- Designed embedded firmware for synchronized data logging from GPS, IMU, temperature, humidity, pressure, and gas sensors.
- Implemented fault-tolerant sensor polling, SD-card logging, and real-time telemetry transmission.
- Analyzed collected atmospheric data to study altitude-dependent variations in environmental parameters.

8-Wheeled Autonomous Rover Project Demo

- Designed and developed an 8-wheeled autonomous ground rover for unstructured-terrain navigation.
- Implemented ROS2-based navigation integrating GPS, IMU, and wheel odometry for localization.
- Developed real-time obstacle avoidance using depth sensing and sensor-fusion algorithms.
- Deployed perception and planning modules on Jetson Nano for onboard autonomous decision-making.

CubeSat Prototype Project Demo

- Built a functional CubeSat prototype integrating power, data acquisition, and onboard processing subsystems.
- Programmed microcontrollers in C++ for synchronized sensor-data logging and telemetry simulation.

Mission CHANDRA Rover Project Demo

- Developed electronics and object-detection systems using Arduino Nano, NodeMCU, and Raspberry Pi Zero.
- Won the state-level SAC-ISRO rover challenge among 100+ teams.

Maze Solver Bot Project Demo

- Programmed an autonomous robot on Raspberry Pi Pico achieving 90% maze-solving accuracy.
- Implemented a custom navigation algorithm using ToF sensor input.

Publications & Intellectual Property

- **Patent Published:** "Eight Wheeled Rover Robot," published 25 June 2026.
- **Review Paper:** "Doppler Limits of LoRa in LEO: A Critical Review of the First In-Orbit Flight Tests," published in *Recent Trends in Electronics and Computer Science (RTECS)*, Volume 12, Issue 02, Pages 21–27, 2025. Read it here

Education

Birla Vishvakarma Mahavidyalaya Engineering College, B.Tech. in Electronics and Communication Engineering — Sept 2022 – 2026

- **CPI: 9.09/10.0**
- **Gold Medalist: Highest CPI, EC Department**

Certifications & Achievements

- **Student of the Year 2026** – BVM Engineering College 2026
- **Gold Medalist** – Highest CPI in 4th Year, BVM 2026
- **Gaurav Puraskar Winner** – BVM 2025

- **1st Place, Robofest 4.0** – National Level Robotics Competition, Rover Category 2025
- **Gaurav Puraskar Winner** – BVM 2024
- **3rd Prize, ECE Project Expo** – BVM, CubeSat Prototype 2024
- **1st Place, Mission CHANDRA** – State Level SAC-ISRO Rover Design Challenge 2023
- **2nd Round Qualifier, Robofest 3.0** 2023
- **Silver Medalist** – Highest CPI in 1st, 2nd & 3rd Year, EC Department, BVM 2025